HAMNA ARSHAD

Software Developer | Full-Stack | Python

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Profile

Computer Science undergraduate at UET Lahore with hands-on experience in **Python, full-stack development, backend development, databases, REST APIs, and AI-powered applications**. Experienced in building applications using **Next.js, React, TypeScript, Python, Flask, PostgreSQL, Supabase, and SQL**. Worked on full-stack projects involving document processing, APIs, database systems, and retrieval-based AI applications. Strong foundation in **OOP, data structures, problem solving, and database development**, with an interest in building practical software products.

Skills

Languages
- Python
- C++
- C#
- JavaScript
- TypeScript

Backend & APIs
- Flask
- REST APIs
- API Integration

Web & Software Development
- React.js
- Next.js
- Flask
- HTML
- CSS
- Tailwind CSS
- Bootstrap
- .NET

AI / Application Integration
- OpenAI API
- Gemini API
- RAG
- Embeddings

Tools & Platforms
- Git
- GitHub
- Google Colab
- Odoo

DataBase Management
- PostgreSQL
- SQL
- MongoDB
- Supabase
- pgvector

Programming Concepts
- Object-Oriented Programming
- Data Structures & Algorithms
- Problem Solving
- Database Management
- Data Modeling

Education

BS Computer Science, *University Of Engineering And Technology* 2023 – 2027
Continue

FSC Pre-medical (with additional Maths), *Aspire College*

Matriculation
92% Marks, Honours kindergarten

PROJECTS

Conversa — AI Chatbot Builder, *Next.js, TypeScript, Python, Supabase, PostgreSQL, pgvector, OpenAI*
- Worked on a full-stack chatbot builder that allows businesses to create chatbots from their own documents and web content.
- Worked with document and web-content extraction, processing, and semantic chunking.
- Implemented a human-in-the-loop workflow for reviewing and validating extracted content before it is used by the chatbot.
- Worked with PostgreSQL and **pgvector** for storing and retrieving document embeddings.
- Implemented keyword search and hybrid retrieval alongside vector-based search.
- Integrated retrieved content with an LLM-based response generation pipeline.
- Worked across frontend, backend, database, authentication, and AI/RAG components.

AutoSlides — AI Research Proposal to PowerPoint Generator,
Python, Flask, Gemini API, Document Processing
- Developed an application that converts research proposal documents into structured PowerPoint presentations.
- Built a Flask backend to process requests and coordinate the application workflow.
- Integrated the Gemini API for AI-assisted content processing.
- Worked with document processing and automated presentation generation.
- Designed a structured workflow for transforming source documents into presentation content.

Database Management System, *Database Architecture, Graph Database, Cypher-like Query Processing*
- Developed a graph-based database management system.
- Worked on data modeling and database architecture.
- Implemented a custom Cypher-like query processor.
- Worked on query execution and graph-based data operations.

WhatsApp Clone, *HTML, CSS, Bootstrap, JavaScript*
- Developed a WhatsApp-inspired interface and messaging simulation.
- Used data structures including **trees and hash maps** to simulate chat functionality and message threading.

Windows Forms Spotify Application, *C#, .NET, SQL, OOP*
- Developed a Spotify-inspired desktop application using C# Windows Forms.
- Implemented playlist, track, and user interaction functionality.
- Used SQL for data storage and management.
- Applied object-oriented programming principles in application development.

Event Management Application, *C++, SQL*
- Developed a console-based event management application.
- Implemented event-related functionality using C++.
- Integrated SQL-based data management.

Snow Bro Game, *C++, SQL*
- Developed a console-based game using C++.
- Implemented game functionality and logic.
- Used SQL for data-related functionality.

Solitaire Game, *C#*
- Developed a Solitaire game using C#.
- Applied data structures including **stacks and queues** for card movement and game logic.

Certificates

- Introduction to Database and SQL
- MongoDB
- Odoo Web Development

Languages

- English
- Urdu